

LALITH KUMAR GUDIPUDI

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EDUCATION

Montclair State University, NJ, USA

August 2025 - May 2027

Masters in Computer Science

Vidya Jyothi Institute of Technology Hyderabad, India

November 2021 - June 2025

Bachelor of Technology in Computer Science Engineering

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

April 2026 - Present

- Conducting research in AI-driven Augmented Reality (AR) security and privacy, focusing on sensor data analysis, adversarial machine learning, and robustness of AR perception systems.
- Developing and evaluating adversarial optimization techniques using PointNet++, Particle Swarm Optimization (PSO), and CMA-ES to analyze vulnerabilities in 3D perception and recognition models.
- Building real-time multimodal sensor streaming and data-processing pipelines using HL2SS and Microsoft HoloLens 2 for AR experimentation, data analysis, and security evaluation.
- Conducting literature reviews and experimental analysis to study adversarial attacks on AR/3D perception systems, supporting the design and evaluation of security-focused research experiments.

PROJECTS

AUTHENT-X: Blockchain-Powered Anti-Counterfeit System

- Developed a blockchain-based anti-counterfeit system that stores product information and authentication records as tamper-resistant digital signatures.
- Implemented QR-code authentication for real-time product verification, enabling users to validate product authenticity across physical and online transactions.
- Designed a decentralized proof-of-concept platform to improve supply-chain transparency, data integrity, and resistance to counterfeit product fraud.

SmartArmor – Secure IoT Data Sharing with Blockchain and Proxy Re-Encryption

- Developed a secure IoT data-sharing system using Proxy Re-Encryption (PRE) to protect data shared between users and edge devices.
- Used Information-Centric Networking (ICN) to improve data distribution and reduce communication overhead.
- Integrated blockchain to provide decentralized, tamper-resistant data sharing and tested the system for security and reliability.

Eye Ball Cursor – Assistive Human-Computer Interaction using OpenCV

- Developed a vision-based system to replace traditional mouse input with eye-tracking, enabling hands-free computer control for users with disabilities.
- Implemented real-time face, eye, and pupil detection using OpenCV and Dlib, mapping gaze direction to cursor movement, and blinks to click operations.
- Designed the solution as an accessible, low-cost assistive technology with applications in healthcare, accessibility tools, gaming, and VR/AR interaction.

SKILLS

Programming Languages

Python, Java, JavaScript, R, C, PHP, HTML

AI / ML

Machine Learning, Computer Vision, Adversarial Machine Learning, PointNet++

Data / Databases

NumPy, Pandas, SQL, MySQL, MongoDB

Cloud / DevOps

AWS, Microsoft Azure

Tools / Technologies

HL2SS, OpenCV, Dlib, GitHub, Jupyter, VS Code, Microsoft HoloLens 2, Tableau